

STATISTICS MASTER'S COMPREHENSIVE EXAMINATION

25 March 2023

Instructions: Attempt any five out of the following six questions. Specify which five you would like to have graded. Students are allowed calculators, laptops, books and notes. Internet use and cell phones are prohibited. Students are allowed to use laptops for viewing electronic copies of textbooks, but for nothing else.

Problem 1. The State of California has a state lottery game called Lotto. To play, you select a 6 set from $\{1, 2, \dots, 49\}$. There are various gadgets sold to generate a six-element subset of $\{1, 2, \dots, 49\}$; one such gadget is pictured below. There are 49 numbered holes and six balls enclosed by a plastic cover. One shakes the balls around and uses the six set determined by their final resting place.

PICK 6 LOTTO & WIN								
1	2	3	4	5	6	7	8	9
○	○	○	○	○	○	○	○	○
10	11	12	13		14	15	16	17
○	○	○	○		○	○	○	○
18	19	20	21		22	23	24	25
○	○	○	○		○	○	○	○
26	27	28	29		30	31	32	33
○	○	○	○		○	○	○	○
34	35	36	37		38	39	40	41
○	○	○	○		○	○	○	○
42	43	44	45		46	47	48	49
○	○	○	○		○	○	○	○

We wish to test if such gadget indeed lead to uniformly-distributed results. To test this, 100 trials were performed. The gadget was vigorously shaken and set down on a flat surface. Following each six set is X – the number of balls falling on the outer perimeter in that 6-set. For example, the first 6-set $\{10, 11, 13, 25, 36, 42\}$ had 3 outside numbers, 10, 25, 42, so $X = 3$. There are 25 outside numbers out of 49. If the six sets were chosen uniformly at random, X would have a hypergeometric distribution $\mathbb{H}(X = j) = \binom{25}{j} \binom{24}{6-j} / \binom{49}{6}$. These numbers and also the empirical frequencies are given in the table below.

Hypergeometric and Empirical Probabilities for X .							
j	0	1	2	3	4	5	6
$H\{X = j\}$	.013	.091	.250	.333	.228	.016	.010
Empirical	.01	.04	.14	.35	.30	.13	.03

- (1) Carry out a Pearson's chi-squared test, report the test-statistic and its p -value. You should have with you a text book with the needed chi-squared table.

Answer: The test-statistics equals 10.23 on chi-squared distribution with six degrees of freedom, with p -value slightly larger than 10%.

- (2) Some statistician suggests to use the average, $\bar{X}$, as a statistic. Under the null distribution, $E(\bar{X}) = 3.06$, $SD(\bar{X}) = 0.116$. Calculate the observed $\bar{X}$, and the z -value.

Answer: The observed $\bar{X} = 3.40$. This yields a standardized (z value) of 2.92.

- (3) Do you think this gadget outputs uniformly-distributed results?

Answer: The chi-squared test fails to reject the null-hypothesis, while the z value shows convincing evidence against uniformity. Since chi-squared test only works well for large sample sizes, the second test is more convincing.

Problem 2. Consider the model $y_i = \beta_0 + x_i\beta_1 + \varepsilon_i$ where ε_i are independent $N(0, \sigma^2)$ for $i = 1, \dots, 5$. We have the following data:

X	Y
-5	-2
-3	1
0	4
2	6
1	3

- (1) Provide the estimate of β_0 and β_1 . Draw a scatter plot, including the fitted regression line.

Answer: We can calculate $\bar{x} = 1, \bar{y} = 2.4, s_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = 34, s_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 34$. Therefore $\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} = \frac{34}{34} = 1$ and $\hat{\beta}_0 = 3.4$.

- (2) For each x_i , compute the fitted value of y_i using $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

Answer: The regression predictions for the training data are: -1.6, 0.4, 3.4, 5.4, 4.4.

- (3) Find the residual $y_i - \hat{y}_i$ for $i = 1, 2, 3, 4, 5$. Draw a residual plot.

Answer: Residuals are -0.4, 0.6, 0.6, 0.6, -1.4.

- (4) Calculate R-squared.

Answer: Calculate $r^2 \approx 0.91$.

Problem 3. A box contains 3 yellow balls and 2 red balls. We must choose 2 balls, at random, from the box. First we flip a fair coin. If it comes up heads we sample with replacement; otherwise we sample without replacement. Let the r.v. Y = the total number of red balls we get.

- (1) What is the distribution of Y ?

Answer: $P(Y = 0) = P(\text{head and } Y = 0) + P(\text{tail and } Y = 0) = 1/2 \times 3/5 \times 3/5 + 1/2 \times 3/5 \times 2/4 = 33/100, P(Y = 2) = P(\text{head and } Y = 2) + P(\text{tail and } Y = 2) = 1/2 \times 2/5 \times 2/5 + 1/2 \times 2/5 \times 1/4 = 13/100, P(Y = 1) = 1 - P(Y = 0) - P(Y = 2) = 54/100$.

- (2) Find EY .

Answer: $EY = 1 \times P(Y = 1) + 2 \times P(Y = 2) = 80/100 = 0.8$.

- (3) Find the probability that the first ball chosen was red given that the second ball chosen was red.

Answer: $P(1st\ Red|2nd\ Red) = P(1st\ Red\ and\ 2nd\ Red)/P(2nd\ Red) = P(Y = 2)/P(2nd\ Red) = (13/100)/(2/5) = 13/40$ as all balls have an equal chance of being 2nd.

- (4) Find the conditional probability that the sampling was done with replacement given that $Y = 2$.

Answer: $P(head|Y = 2) = P(head\ and\ Y = 2)/P(Y = 2) = (1/2 \times 2/5 \times 2/5)/(13/100) = 8/13$.

Problem 4. Let X and Y be independent random variables from two different negative exponential distributions with parameters $\theta_1 > 0$ and $\theta_2 > 0$ so their joint density function is

$$f(x, y|\theta_1, \theta_2) = \theta_1 e^{-\theta_1 x} \theta_2 e^{-\theta_2 y} \text{ for } x > 0, y > 0.$$

- (1) Find the form of the most powerful test of

$$H_0 : \theta_1 = \theta_2 = 2 \text{ vs. } H_1 : \theta_1 = 1, \theta_2 = 3.$$

Answer: By Neyman - Pearson, reject H_0 if $f(x, y|1, 3)/f(x, y|2, 2) = (e^{-x} 3e^{-3y})/(2e^{-2x} 2e^{-2y}) = (3/2)e^{x-y} > c$. This is equivalent to rejecting for large $x - y$.

- (2) Consider testing

$$H_0 : \theta_1 = \theta_2 \text{ vs. } H_1 : \theta_1 \neq \theta_2.$$

Find the form of the likelihood ratio test and write it as a function of $x/(x + y)$.

Answer: In general, $\theta^m e^{-u\theta}$ is maximized when $\theta = m/u$ and the maximum will be $(m/u)^m e^{-m}$. Under H_0 the joint density is

$$\theta_1 e^{-x\theta_1} \theta_1 e^{-y\theta_1} = \theta_1^2 e^{-(x+y)\theta_1}.$$

Thus the max under H_0 occurs when $\theta_1 = 2/(x + y)$ and is equal $(2/(x + y))^2 e^{-2}$. Under H_1 the joint density is $\theta_1 e^{-x\theta_1} \theta_2 e^{-y\theta_2}$. Thus the max under H_1 occurs when $\theta_1 = 1/x$, $\theta_2 = 1/y$ and is equal $(\frac{1}{xy}) e^{-2}$. The LRT will reject for small values of the ratio of max H_0 / max H_1 . Equivalently, large values of

$$\frac{xy}{(x + y)^2} = \frac{x}{x + y} \left(1 - \frac{x}{x + y}\right) = \left(\frac{x}{x + y} - 1/2\right)^2.$$

Problem 5. Suppose that the random variables $Y_1, \dots, Y_n$ satisfy

$$Y_i = \beta x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where $x_1, \dots, x_n$ are fixed constants, and $\varepsilon_1, \dots, \varepsilon_n$ are iid $\mathcal{N}(0, \sigma^2)$, σ^2 unknown.

- (1) Find a two-dimensional sufficient statistic for (β, σ^2) .

Answer: We start by writing the likelihood $L(\theta | \mathbf{y})$, here $\mathbf{y}$ is shorthand for the full data.

$$\begin{aligned} L(\theta | \mathbf{y}) &= \prod_i \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} (y_i - \beta x_i)^2\right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_i (y_i^2 - 2\beta x_i y_i + \beta^2 x_i^2)\right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\beta^2 \sum_i x_i^2}{2\sigma^2}\right) \exp\left(-\frac{1}{2\sigma^2} \sum_i y_i^2 + \frac{\beta}{\sigma^2} \sum_i x_i y_i\right). \end{aligned}$$

By sufficiency principle (see Theorem 6.2.2 of Casella Berger for instance), $(\sum_i Y_i^2, \sum_i x_i Y_i)$ is a sufficient statistic for (β, σ^2) .

- (2) Find the MLE of β , and show that it is an unbiased estimator of β .

Answer:

$$\log L(\beta, \sigma^2 | \mathbf{y}) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_i y_i^2 + \frac{\beta}{\sigma^2} \sum_i x_i y_i - \frac{\beta^2}{2\sigma^2} \sum_i x_i^2.$$

For a fixed value of σ^2 ,

$$\frac{\partial \log L}{\partial \beta} = \frac{1}{\sigma^2} \sum_i x_i y_i - \frac{\beta}{\sigma^2} \sum_i x_i^2 \stackrel{\text{set}}{=} 0 \Rightarrow \hat{\beta} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.$$

Also,

$$\frac{\partial^2 \log L}{\partial \beta^2} = \frac{1}{\sigma^2} \sum_i x_i^2 < 0,$$

so it is a maximum. Because $\hat{\beta}$ does not depend on σ^2 , it is the MLE. And $\hat{\beta}$ is unbiased because

$$\mathbb{E}\hat{\beta} = \frac{\sum_i x_i \mathbb{E}Y_i}{\sum_i x_i^2} = \frac{\sum_i x_i \cdot \beta x_i}{\sum_i x_i^2} = \beta.$$

- (3) Find the distribution of the MLE of β .

Answer: $\hat{\beta} = \sum_i a_i Y_i$, where $a_i = x_i / \sum_j x_j^2$ are constants. Since linear functions of Gaussian random variables are Gaussian, $\hat{\beta}$ is normally distributed with mean β , and

$$\text{Var } \hat{\beta} = \sum_i a_i^2 \text{Var } Y_i = \sum_i \left(\frac{x_i}{\sum_j x_j^2}\right)^2 \sigma^2 = \frac{\sum_i x_i^2}{\left(\sum_j x_j^2\right)^2} \sigma^2 = \frac{\sigma^2}{\sum_i x_i^2}.$$

Problem 6. A group of doctors decided to run a trial to compare three weight loss programs and participants are randomly assigned to one of the comparison programs and are counseled on the details of the assigned program. Participants follow the assigned program for 8 weeks. The outcome of interest is weight loss, defined as the difference in weight measured at the

start of the study (baseline) and weight measured at the end of the study (8 weeks), measured in pounds. Three popular weight loss programs are considered. The first is a low calorie diet. The second is a low fat diet and the third is a low carbohydrate diet. For comparison purposes, a fourth group is considered as a control group. Participants in the fourth group are told that they are participating in a study of healthy behaviors with weight loss only one component of interest. The control group is included here to assess the placebo effect (i.e., weight loss due to simply participating in the study). Patients agreeing to participate in the study are randomly assigned to one of the four diet groups. Weights are measured at baseline and patients are counseled on the proper implementation of the assigned diet (with the exception of the control group). After 8 weeks, each patient's weight is again measured and the difference in weights is computed by subtracting the 8 week weight from the baseline weight. Positive differences indicate weight losses and negative differences indicate weight gains. For interpretation purposes, we refer to the differences in weights as weight losses and the observed weight losses are shown below.

Low Calorie	Low Fat	Low Carbohydrate	Control
8	2	3	2
9	4	5	2
6	3	4	-1
7	5	2	0

- (1) We want to check if there a statistically significant difference in the mean weight loss among the four diets. Write down an appropriate hypothesis (null and alternative hypothesis) for it.

Answer: Let $\mu_1, \mu_2, \mu_3, \mu_4$ respectively denote the means of the low calorie, low fat, low carbohydrate, and control group. Then we would like to perform the hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$, $H_1 : \text{Not all means are equal}$.

- (2) Write down the ANOVA table. You may use that the total sum of squares is 114.4.

Answer: We already know that total sum of squares is 114.4. Note that total sum of squares is the between treatment sum of squares plus residuals (within group sum of squares). Group means are 7.50, 3.50, 3.50, and 0.75. The overall mean is 3.1825. The between sum of squares is 92.7. Calculate

- (3) Do you accept your null hypothesis?

Answer: The critical value for a level 0.05 test is 3.49. We reject the null hypothesis as the F-statistic 8.54 > 3.49.

<i>Source of Variation</i>	<i>Sums of Squares (SS)</i>	<i>Degrees of Freedom (df)</i>	<i>Means of Squares (MS)</i>	<i>F</i>
<i>Between Treatment</i>	<i>92.7</i>	$4 - 1 = 3$	$92.7/3 = 30.9$	$30.9/1.81 = 17.05$
<i>Error (or Residual)</i>	<i>21.8</i>	$16 - 4 = 12$	$21.8/12 = 1.81$	
<i>Total</i>	<i>114.4</i>	$16 - 1 = 15$		